

ELLIOT, South Africa

WMO: 686510

Lat: 31.3364S

Lon: 27.8403E

Elev: 4800

StdP: 12.32

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	29.8	7.4	9.6	59.0	11.2	11.6	57.2	24.8	58.5	22.5	59.9	1.1	300	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.2	86.6	60.1	83.8	59.8	81.0	59.0	66.6	75.8	65.4	74.5	64.1	73.3	10.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	106.4	70.7	62.4	101.0	69.3	61.1	96.4	67.9	34.3	75.6	33.1	74.6	32.0	73.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.8	14.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	65.8	65.9	63.6	58.3	54.7	49.2	49.1	51.9	56.4	58.8	60.7	63.9
	DBStd	8.62	5.48	5.23	5.53	6.06	6.46	6.06	6.76	7.28	7.45	7.33	6.75	5.79
	HDD50	332	0	0	0	10	25	87	99	64	25	15	7	0
	HDD65	2837	56	46	92	209	326	473	492	408	269	213	160	93
	CDD50	3311	490	444	421	259	171	64	72	124	218	289	327	432
	CDD65	342	81	71	48	8	7	0	0	3	13	22	29	60
	CDH74	3243	628	486	371	87	54	0	0	32	227	403	357	598
	CDH80	767	177	125	62	7	12	0	0	0	37	88	80	179
(14) Wind	WSAvg	6.4	6.4	5.9	5.5	4.7	5.7	7.1	6.8	6.9	6.9	7.1	7.2	7.1
(15) Precipitation	PrecAvg	27.8	4.6	3.9	3.7	2.1	0.7	0.6	0.6	0.9	1.3	2.4	3.3	3.7
	PrecMax	42.6	9.5	6.5	7.0	3.8	2.3	2.4	2.6	3.8	4.4	8.3	8.0	6.7
	PrecMin	8.7	1.9	1.2	1.1	0.5	0.0	0.1	0.0	0.0	0.1	0.4	0.4	1.1
	PrecStd	6.9	1.8	1.5	1.5	0.8	0.6	0.7	0.7	0.9	1.1	1.7	1.9	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.5	87.9	85.3	81.1	80.9	68.5	70.2	78.6	84.1	87.0	87.2	89.7
		MCWB	62.0	62.2	60.6	59.2	60.3	48.4	46.5	51.1	54.0	56.5	57.9	59.1
	2%	DB	85.0	84.3	81.4	76.6	74.2	65.7	66.9	74.0	79.9	82.5	82.3	84.9
		MCWB	62.1	62.6	60.3	56.5	55.2	46.6	46.1	49.5	51.9	55.5	58.2	60.2
	5%	DB	80.8	80.4	78.2	73.5	71.2	63.5	64.4	70.4	76.0	78.9	78.4	80.8
		MCWB	62.5	62.1	60.1	55.6	52.5	45.6	45.3	48.2	51.2	55.0	57.3	60.6
	10%	DB	76.8	76.2	75.0	70.2	67.7	61.0	61.7	66.9	72.1	74.7	74.4	76.4
		MCWB	62.3	62.1	59.2	55.2	50.4	44.7	44.7	47.2	50.7	53.8	57.6	61.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	68.4	66.8	62.9	65.5	51.7	49.7	54.8	58.9	61.6	63.7	68.0
		MCDB	79.0	76.3	75.4	74.4	74.7	63.0	60.9	67.4	71.5	75.0	74.2	76.1
	2%	WB	66.4	66.4	64.9	60.7	59.1	49.4	48.4	52.4	56.4	59.8	61.9	65.4
		MCDB	76.2	75.1	74.3	70.0	72.8	60.3	61.1	66.7	70.2	73.5	72.1	74.8
	5%	WB	64.9	65.0	63.3	59.0	54.4	47.9	47.0	50.6	54.7	58.0	60.6	63.8
		MCDB	74.1	73.8	72.6	68.6	67.2	58.7	60.3	65.3	68.3	71.3	71.1	73.1
	10%	WB	63.6	63.6	61.7	57.7	52.2	46.5	45.9	48.8	53.1	56.5	59.3	62.3
		MCDB	72.7	72.3	70.6	66.7	64.7	57.6	59.2	63.0	65.7	69.2	70.4	71.6
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	18.1	19.4	20.5	22.8	21.8	23.3	24.4	25.2	23.2	20.3	20.1
		MCDBR	26.4	26.4	25.7	26.3	27.1	24.2	25.6	28.3	32.1	32.7	29.9	30.1
		MCWBR	8.8	8.7	8.9	10.7	10.5	10.9	10.6	11.3	10.5	10.6	9.0	8.8
	5% WB	MCDBR	20.2	20.2	20.1	19.8	23.8	22.0	21.4	25.2	25.5	25.7	22.1	20.9
		MCWBR	10.0	9.4	9.1	9.2	10.9	11.5	10.0	11.1	10.4	10.6	9.5	10.2
(40) Clear-Sky Solar Irradiance	taub		0.336	0.338	0.323	0.292	0.272	0.256	0.257	0.306	0.341	0.349	0.326	0.330
	taud		2.484	2.478	2.496	2.563	2.589	2.600	2.583	2.429	2.330	2.352	2.471	2.496
	Ebn at Noon		318	312	306	300	292	290	295	290	294	304	319	321
	Edh at Noon		37	36	33	28	24	22	23	31	38	40	37	37
(44) All-Sky Solar Radiation	RadAvg		2120	1938	1686	1384	1137	1012	1104	1330	1660	1889	2122	2194
	RadStd		142	156	115	75	56	60	61	86	98	138	126	131

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page